

**NUIMPACT**

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# Sector Spotlight Digest



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Northeastern University





## Table of Contents

| Title                                                                              | Page |
|------------------------------------------------------------------------------------|------|
| <b>NUImpact Overview</b>                                                           | 3    |
| <b>Education &amp; Financial Services: ESG Reporting Software</b>                  | 4    |
| <b>Energy &amp; Environment: Water Infrastructure Improvement</b>                  | 5    |
| <b>Food &amp; Agriculture: Robotic Automation</b>                                  | 6    |
| <b>Healthcare: Nutritional Care</b>                                                | 7    |
| <b>Technology, Media, &amp; Telecommunication: Digital Accessibility Platforms</b> | 8    |
| <b>Sources</b>                                                                     | 9    |



## About NUImpact

- NUImpact is Northeastern University's student-led impact investing initiative and investment fund.
- Since our inception in Spring 2016, the organization has served as a unique resource and thought exchange for the Northeastern community to understand purposeful capital, develop technical skills, and make targeted investments in companies that promote positive social and environmental impact
- Our three-pronged investment thesis seeks investment in companies that are:

**Northeast-Focused**

**Aid Underserved Communities**

**Financially Sustainable**

## Our Research Arm Yields Significant Benefits

- The research analyst program places a strong emphasis on industry research to enhance our fund's impact diligence process.
- A key responsibility for research analysts is crafting concise one-page reports spotlighting a chosen sector within their verticals.
- Some benefits of this new program include:

### Informed Decision Making

Offers the fund with valuable insights into specific industries, enabling risk mitigation and better-informed investment decisions

### Refined Impact Assessment

Enhanced sector understanding equips analysts with knowledge needed to conduct robust impact diligence and measurement

## Industry Focus



**Education & Financial Services**



**Energy & Environment**



**Food & Agriculture**



**Healthcare**



**Technology, Media, & Telecommunication**

## Meet the Research Team



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## Industry Overview

- Environmental, Social, Governance (ESG) focuses on promoting operations that **reduce companies’ environmental footprint** and **establish sustainable business practices**.
- Large firms, such as, Deloitte, KPMG, and PwC offer **general reporting** while smaller, specialized firms like Green Buoy consulting offer **catered help**.
- These services provide support with data collection services for **easy implementation** with most common customers being **large and middle market producers**

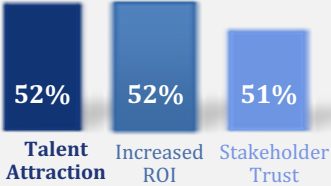
## Favorable Tailwinds

### Investor Demand

Companies are incentivized to pursue ESG measures to **appease** potential and current **shareholders**

**Complicated** and **expensive** reporting processes create **reliance** on ESG specialized firms

- Expected **benefits** of **ESG reporting** as represented by 300 companies



### Boston’s Green New Deal

Investment Funds, such as the Community Bank Investment Initiative, incentivize ESG practices

Promoting ESG benefits and practices that are standardized within Boston public schools

## S&P 500 Reporting Framework

- Frameworks establish a **set standard** for ESG reporting companies to compartmentalize procedures.

### 2022 Companies Reporting



## Emerging Headwinds

### Reporting Inaccuracies

Companies are incentivized to **over-report** ESG contributions, particularly with **Scope-3** emissions

Various **frameworks** and **reporting strategies** create **inconsistent** or hard-to-find data

### Cost of Reporting

- Companies under financial stress may be disincentivized to pursue ESG reporting measures

**\$250,000 – \$450,000**

Average cost spent on ESG reporting



## Recent Deals Highlights



Acquired by LexisNexus

Utilizes Intelligize platforms to expand analytical reporting and commercial data findings for compliance-based clients



Raised \$18 Million

Raised capital in June 2024 for global expansion and greater outreach with natural disaster protection and prediction software



Acquired Hg

Intends to enhance Cubes compliance offering, making digital reporting accessible to larger and smaller firms

Industry Overview

13.9% CAGR

\$107.3B by 2030

- Enhances water safety and efficiency with modern technologies to address the dated, crumbling systems in place today.
- Low-income communities are disproportionately affected by water quality violations, with 70% of impacted U.S. residents residing in areas of highest social vulnerability.

Industry Tailwinds

Bipartisan Unfractured Law (Nov 2021)

\$50M

Allocated in **principle forgiveness loans** towards improving water infrastructure, the **largest investment** in US history

State-Specific Initiatives

- Growing climate challenges and public health concerns has escalated state-specific water infrastructure improvement focus..
- 

Mass: **Healey-Discroll Administration** awards **\$1.4 billion** in loans
- 

NY: **Water Infrastructure Improvement Act** awards **\$450 million** in 2023 alone
- 

NH: **Lead Service Line Inventory Act** allows for **\$100,000 grants** in eligible communities

Extreme Weather Accelerating Investment Needs

- Major funding will be needed to maintain water infrastructure with anticipated climate affects.

\$625B over 20 years

to maintain drinking water infrastructure

\$448B - \$944B by 2050

to adapt systems for climate change consequences

Technological Drivers

- Smart Sensors:

Monitor water quality, leaks, and system pressure in real-time
- Ai-Driven Predictive Maintenance:

Helps identify and preemptively repair infrastructure failures.
- Bio-filtration:

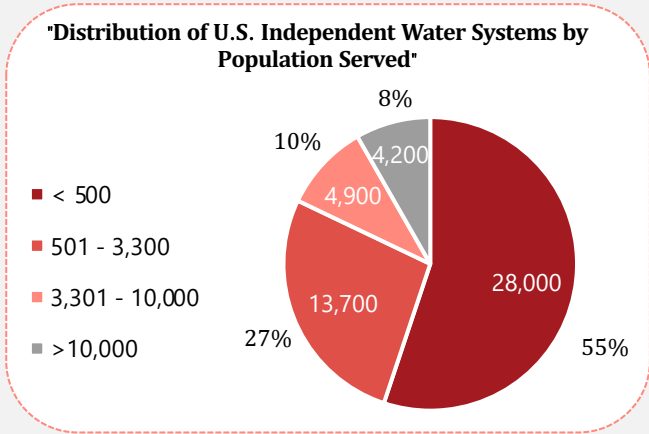
Use of natural microorganisms to treat water for organic pollutants

Industry Headwinds

- Aging Infrastructure

  - Deteriorating systems increase the complexity and cost of upgrades, creating financial and operational challenges for implementing modern, resilient solutions.
- Fragmented Market

  - Across the United States, there are more than 51,000 separate and independent water systems.
  - Over half (55%) of these systems serve communities of 500 people or fewer, making widespread upgrades extremely challenging.



Recent Deal Highlights

EDGE AI

- Boston-based Edge AI leverages AI technology to deliver predictive analytics and monitoring solutions for water infrastructure systems.
- In October 2024, **Edge AI** successfully raised a **\$3M seed round**, with primary backing from LMNT Ventures and the Organic Growth Fund.



- Aqua Membranes uses 3D printing to create efficient, durable water filtration membranes for municipal and industrial use.
- In October 2024, it was announced that **Aqua Membranes** had closed an **undisclosed funding** round led by Burnt Island Ventures.





## Industry Overview & Operational Impact

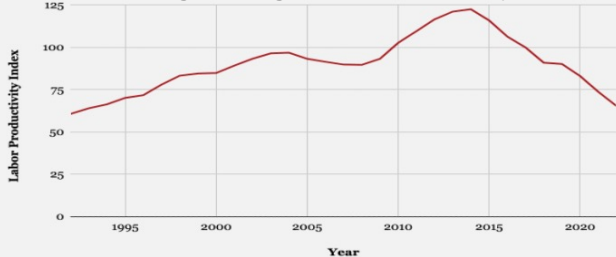
- **Farmers and food handlers must balance productivity with conservation** to support **growing food demand while protecting the environment** in their operations

34.5% CAGR

\$20.3B Market Size by 2025

- To address nuances in smaller farms/food providers, startups have been shifting business models to **work more closely with customers**

### Warehousing & Storage Labor Productivity (U.S. BLS)



## Social & Stakeholder Impact

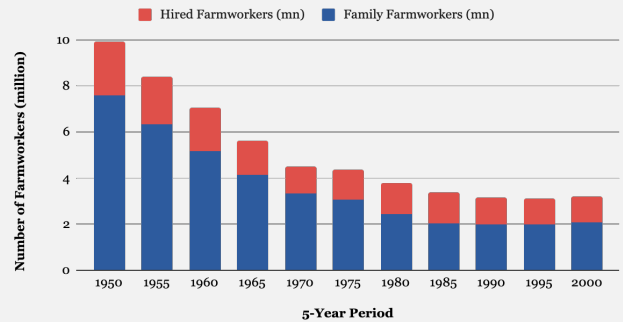
### Robotic Automation Improves:

Food and farm workers' standard of labor

Reduces operational labor shortages

Sustainability through resource conservation

### Change in Farm Owners' Relations to Workers (USDA)



## Future Outlook

### Tailwinds

To support R&D, sustainability, & implementation of products to end-users, the NSF, NIFA, & USDA have generated funding

Robotic automation can be tailored to address specific farm needs as there's a need for integration with human employees due to current labor shortages

Climate change increases variability, and robotic automation will help stakeholders adapt to changes.

Companies are targeting smaller farms in agriculture to increase access, for both R&D and testing

### Headwinds

With high initial investment costs and limited economies of scale, VCs are notably worried about low adoption rates in agriculture

System integrations with technology might be disproportionately harder for smaller farmers with more diversified operations

Knowledge and training is required, affecting small operations through potentially outdated technology

Robotic systems perform better with access to higher amounts of data, which might require additional expensive investments (sensors)

## Relevant & Model Startups



### Ubiros



### Harvest Automation

Overview

Located in **Natick, MA**, the company innovates **food packaging & order fulfillment**, notably with **Gentle** (their soft robotic gripper)

Impact

An early-stage startup that wants to **reduce operational redundancies**, all while **addressing the labor shortage** in order fulfillment warehouses

Located in **Billerica, MA**, the company specializes in **mobile robots** that work with humans, especially in **greenhouse & fruit/vegetable farming** operations

Tailored solutions to **automate agricultural tasks** and **improve labor quality**, while also **increasing crop yields** with **less pesticides** and herbicides



## Sector Overview

- Nutritional care plans are created by registered dietitians, including **medically tailored meals** and ongoing nutrition education
- The shift towards **value-based care** puts a premium on patient-centric nutritional care programs
- 13.5%** of US households are **food insecure**

## 2024 US Nutritional Care Market Outlook

8.05% CAGR  
(2024 – 2029)

\$106.46B  
Market Size

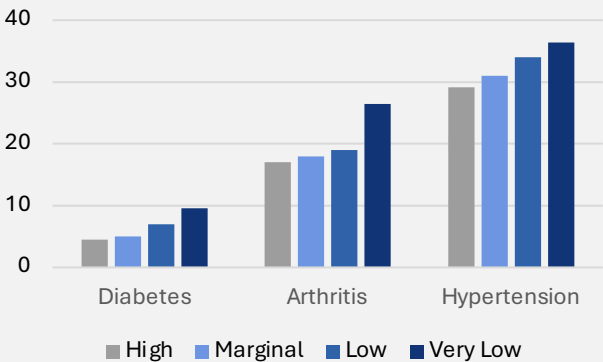
- Food insecurity** costs the US health care system an **\$53 billion** annually

## Future Outlook

### Tailwinds

60% American adults have a nutrition-related chronic disease

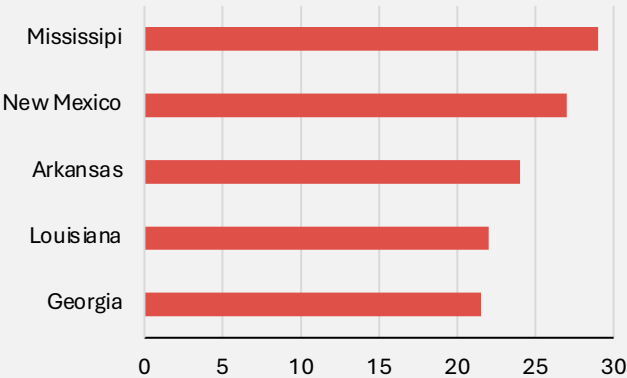
Predicted Disease Prevalence % for Adults in Low-Income Households



### Headwinds

**Food deserts** limit opportunities for underserved communities to access nutritional care programs

Top States by % of Population in Food Deserts



Medical Nutrition Therapy Act of 2023 expands Medicare coverage of nutritional therapy services to **major chronic diseases**

**10** states have not adopted Medicaid expansion, leaving **1.5 million** uninsured people

Medically tailored meals could avoid **1.6 million hospitalizations** and save **\$13.6 billion** annually

**No** comprehensive national legislation: Medicaid-covered medically tailored meals aimed at only select-few chronic conditions

## Relevant Companies



PROJECT BREAD

### Project Bread

Massachusetts-based charity providing individualized meals to hospitals; ~18M revenue



### Sustainable CAPE

Farm-to-table charity partnering with Cape Code healthcare systems; \$413k revenue



The Food Project

### The Food Project

Charity creating sustainable, urban food systems in Massachusetts; ~3M revenue

Sector Overview

- Digital accessibility software encompasses all tools and platforms that assist or enhance the in-person and online experiences of disabled people
- B2B products include enterprise compliance software, while assistive technologies dominate the D2C market
- Legislation such as the Americans with Disabilities Act (ADA) ensures that disabled people have equitable access to technology and digital services

2024 US Digital Accessibility Software Market

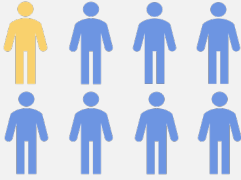
8.4% CAGR  
(2024 – 2031)

\$201.5M  
Market Size  
by 2030

- In July 2023, Massachusetts Governor Maura Healey established the Digital Accessibility and Equity Governance Board

Favorable Tailwinds

- Approximately **1 in 8** Americans have a disability
- As the population ages, the prevalence of hearing and vision disabilities will increase



Advancements in **AI, ML and NLP algorithms** have improved digital accessibility by removing communication barriers

Emerging Headwinds

- **83%** of educators did not receive assistive technology training while completing their college degree

Accessibility software is expensive, ranging from **\$700 to over \$10,000 in TCO**

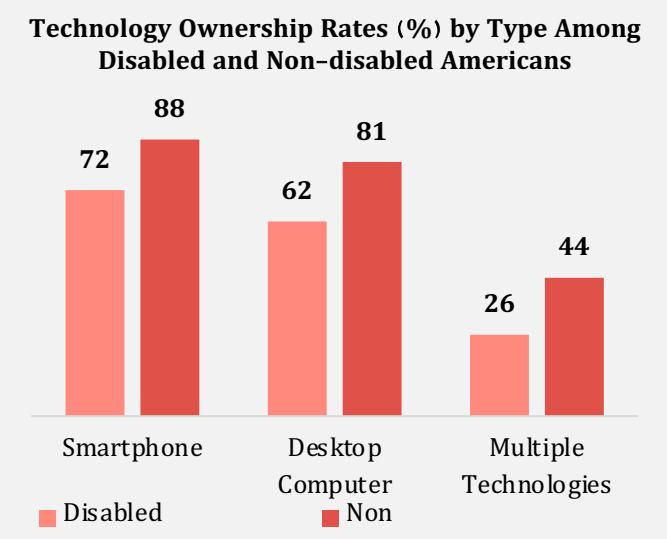
AI and NLP-based Assistive Technologies

Summarization & Captioning

Alternative Text Visualization

Speech Generation & Recognition

30% of students with disabilities use assistive technologies in the classroom; assistive technologies range from **special tablets to screen reading software**



Recent Deal Highlights

Ahoi!

- Based in Boston, MA, **Ahoi!** is a directory and mapping service that uses community reviews to assess the accessibility of specific locations
- In May 2024, Ahoi! joined Techstars, receiving **\$20,000** in funding; the firm is currently generating revenue and is pursuing contracts with the city of Boston and other municipalities

TestParty

- Founded in New York, NY, **TestParty** is the developer of a B2B compliance software program designed to help firms meet ADA, EAA, and WCAG standards
- In June 2024, TestParty raised **\$4 million** in seed funding from the National Science Foundation, Harlem Capital, and other VC funds





## Education & Finance: ESG Reporting Software

1. *Boston.gov*, “Boston’s Green New Deal” October 2024
2. *CEEC*, “ESG Report Pricing” November 2024
3. *Community*, “LexisNexis Announces Its Intention to Acquire Intelligize – Expanding Its Securities and M&A Offering and Continuing to Invest in Analytical Tools” November 2024
4. *CUBE Global*, “ESG Strategy and ESG Compliance”
5. *Greenly*, “What is ESG Reporting, and Should You Be Doing It?” June 2024
6. *Pitchbook*, “Climate X Company Profile: Valuation and Investors” November 2024
7. *TheCAQ.org* “S&P 500 ESG Reporting and Assurance Analysis.” November 2024

## Energy & Environment: Water Infrastructure Improvement

1. *Mass.gov* “Healey-Driscoll Administration Awards \$1.4 Billion in Loans and Grants to Fund Wastewater and Drinking Water Infrastructure Projects” March 2024
2. *New Hampshire Department of Environmental Services*, “Drinking Water Related Grants” October 2024
3. *New York State*, “Water Infrastructure Improvement & Intermunicipal Grant Awards” December 2023
4. *United States Environmental Protection Agency*, “Small Drinking Water System Variances” February 2024
5. *United States Environmental Protection Agency*, “Water Infrastructure Investments” November 2024

## Food & Agriculture: Robotic Automation

1. *America’s Seed Fund – US Small Business Administration*, “About SBIR and STTR” 2024
2. *MassRobotics*, “Robots in Food Tech” 2021
3. *US Bureau of Labor Statistics* “Graphics for Economic News Releases – Labor Productivity Indexes by Industry” 2024
4. *US National Institute of Food and Agriculture*, “Cyber-Physical Systems (CPS) (Joint NSF) Competitive Grants Program” 2019
5. *US National Science Foundation*, “NSF and USDA Join Forces to Boost Innovation in Agricultural Robotics” 2024
6. *PWR Pack* “Food Industry – How Robots are Used for Food Packaging and Processing” 2022

## Healthcare: Nutritional Care

1. *Pitchbook*
2. *The Food Trust* “Addressing Nutrition and Food Access in Medicaid: Opportunities and Considerations”
3. *USDA Economic Research Services*, “Key Statistics & Food Security Status of U.S. Households” 2023
4. *United States Department of Agriculture*, “Food Insecurity, Chronic Disease, and Health Among Working-Age Adults” 2017

## TMT: Digital Accessibility Platforms

1. *Journal of Special Education Technology*, “Teachers of Students With Learning Disabilities: Assistive Technology Knowledge, Perceptions, Interests, and Barriers.” 2020
2. *KBV Research*, “North America Digital Accessibility Software Market” October 2024
3. *US Census*, “Facts for Features – Disabilities Act” May 2021
4. *Pew Research*, “Americans with Disabilities are Less Likely than Those Without To Own Some Digital Devices” September 2021



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